

medical case study maths

Constants:

$$ke = 0.1 \quad \text{baseline elimination rate per hour} \quad (1)$$

$$Vd = 30 \quad \text{baseline volume of distribution} \quad (2)$$

$$b_{temp} = 0.002 \quad \text{Drug effect on temp} \quad (3)$$

$$c_{wbc} = 0.02 \quad \text{Drug effect on WBC} \quad (4)$$

Healthy ranges:

$$T_{norm} = (36, 38) \quad (5)$$

$$WBC_{norm} = (4, 12) \quad (6)$$

$$C_{max} = 30 \quad (7)$$

Efficiencies:

$$ke_{eff} = ke \times *1 - 0.004 * (age - 50) \quad \text{Elimination efficiency} \quad (8)$$

$$vd_{eff} = vd * \frac{\text{weight}}{70} \quad \text{Distribution efficiency} \quad (9)$$

Infection:

$$x_{n+1} = rx_n(1 - x_n) \quad (10)$$

Homeostasis:

$$\frac{\Delta H_T}{\Delta t} = -H_T^c (T - 37.5^\circ C) \quad (11)$$

$$\frac{\Delta H_{wbc}}{\Delta t} = -H_{wbc}^c (WBC - 7) \quad (12)$$

System:

$$\frac{\Delta C}{\Delta t} = -ke_{eff} \times C + \frac{D_t}{vd_{eff}} \quad (13)$$

$$\frac{\Delta T}{\Delta t} = T_{eff}^* \quad (14)$$